

Supplementary Material: A Differentiable Atari VCS: A Complex, Fully Known Ground Truth for Explainable AI

Anonymous submission

This supplement gives the full proofs of the two theorems stated in the main paper, together with the implementation-effort plot referenced in the Results. It is self-contained: we first restate the soft/hard formulation, then prove the forward-equivalence and temperature-limit results, and finally show the effort timeline.

1 Setup and Notation

The differentiable emulator runs one of two transition maps over the full VCS state $x = (\text{registers}, \text{RAM}, \text{RIOT}, \text{TIA}, \text{frame buffer})$: the bit-exact *hard* map Φ_H and the differentiable *soft* map Φ_S . Both are the per-opcode handler selected by the byte at the program counter (pc). Four primitives appear in the handlers.

One-hot reads. A ROM or RAM read at integer address a is the dot product

$$\text{peek}(r, a) = \mathbf{1}_a^\top r = \sum_i \llbracket i = a \rrbracket r_i = r_a. \quad (1)$$

Softmax dispatch. With a score vector (logits) $\ell \in \mathbb{R}^K$ over the $K = 256$ opcodes, temperature $T > 0$, and softmax weights $w = \text{softmax}(\ell/T)$, the relaxed dispatch output over candidate handler rows V is

$$\text{select}(\ell, V; T) = w^\top V, \quad w_k = \frac{e^{\ell_k/T}}{\sum_j e^{\ell_j/T}}. \quad (2)$$

Sigmoid branch. For a relaxed status flag $f \in [0, 1]$ written as a logit $z = s(2f - 1)$ (sign s for branch-when-set/clear) and sharpness α , the gate and relaxed program counter are

$$g = \sigma(\alpha z), \quad \text{pc}_{\text{soft}} = (1 - g) \text{pc}_{\bar{b}} + g \text{pc}_b. \quad (3)$$

Straight-through estimator. For a soft value with a matching hard value,

$$\text{STE}(\text{soft}, \text{hard}) = \text{soft} + \text{sg}(\text{hard} - \text{soft}), \quad (4)$$

whose forward value is *hard* and whose gradient is that of soft.

In the executed soft step, dispatch uses the saturated (hard) form of (2), and the branch returns $\text{STE}(\text{pc}_{\text{soft}}, \text{pc}_{\text{hard}})$; (2) and (3) in their relaxed form define the “fully relaxed” variant analysed in Theorem 2.

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Assumption (DM, decision margins). Along the executed trajectory of length N , every discrete decision has a strictly positive margin: each dispatch has a logit gap $\Delta = \ell_{k^*} - \max_{k \neq k^*} \ell_k \geq \Delta_{\min} > 0$, and each branch has a margin $m = |z| \geq m_{\min} > 0$.

2 Proofs

Theorem 1 (exact forward equivalence). *For every reachable state x and every finite α, T , $\Phi_S(x) = \Phi_H(x)$; hence $x_t^S = x_t^H$ for all $t \leq N$. The two modes differ only in the Jacobian.*

Proof. It suffices to show that, for each opcode, the soft and hard handlers produce the same output value; the state is then carried forward identically. Each handler is composed of four kinds of primitive.

(i) *Memory reads.* A ROM or RAM read at integer address a is the one-hot dot product (1). Because a is an integer index, $\mathbf{1}_a$ is exact and $\text{peek}(r, a) = r_a$, identical to a hard array read.

(ii) *Arithmetic.* Register, status-flag, binary-coded-decimal (BCD), and RIOT-timer updates use the same integer and round operations as the hard handlers; the soft path stores these quantities as float32 but computes them with integer semantics, so their outputs are equal.

(iii) *Dispatch.* The handler is chosen by a hard switch on the decoded opcode byte, exactly as in hard mode; the softmax form (2) is not used on the forward path.

(iv) *Branch.* A conditional branch forms pc_{soft} from (3) but returns $\text{STE}(\text{pc}_{\text{soft}}, \text{pc}_{\text{hard}})$ (4). As sg is the identity on the forward pass, this equals pc_{hard} for every α and T .

Each primitive’s forward output thus equals the hard handler’s, so $\Phi_S(x) = \Phi_H(x)$ for every reachable x ; the hard handlers are themselves validated bit-for-bit against XITARI by the conformance harnesses. By induction over the N steps of a trajectory, $x_t^S = x_t^H$ for all $t \leq N$.

Bit-exactness. float32 represents every integer in $[0, 2^{24}]$ exactly, and all VCS quantities lie in this range—data bytes are at most 255, addresses are 13 bits, and per-frame cycle counts are a few tens of thousands—so the float32 results are not merely close to the integer results but identical.

Gradients. The stop-gradient in (iv) alters only the reverse-mode rule: it sets $\partial \text{STE} = \partial \text{pc}_{\text{soft}}$, the derivative of the sigmoid-blended counter, which is generically nonzero where the hard branch (a step function of the flag) has zero

or undefined derivative; likewise the read in (i) has Jacobian $\mathbf{1}_a$ with respect to r . The forward pass is unchanged while gradients become available. $\square$

Theorem 2 (temperature-limit bound). *For the fully relaxed variant (softmax dispatch and sigmoid branch, no straight-through correction), under Assumption (DM),*

$$\|\Phi_S^T(x) - \Phi_H(x)\|_\infty \leq C[(K-1)e^{-\Delta_{\min}/T} + e^{-\alpha m_{\min}}],$$

and over N steps with Lipschitz constant $L \geq 1$ the trajectory error is at most $\frac{L^N - 1}{L - 1}$ times this, which is $\mathcal{O}(e^{-c/T})$ with $c = \Delta_{\min}$.

Proof. We bound the per-step deviation, then propagate it. Throughout, C bounds the range of any state value (bytes ≤ 255 , branch offsets ≤ 256).

Dispatch term. At a step with logits ℓ and winner k^* , the gap is $\Delta \geq \Delta_{\min} > 0$ by Assumption (DM). For any loser $k \neq k^*$,

$$w_k = \frac{e^{\ell_k/T}}{\sum_j e^{\ell_j/T}} \leq \frac{e^{\ell_k/T}}{e^{\ell_{k^*}/T}} = e^{(\ell_k - \ell_{k^*})/T} \leq e^{-\Delta_{\min}/T},$$

so the non-winning mass is $1 - w_{k^*} = \sum_{k \neq k^*} w_k \leq (K-1)e^{-\Delta_{\min}/T}$. As the output is the convex combination $\sum_k w_k V_k$,

$$\begin{aligned} \left\| \sum_k w_k V_k - V_{k^*} \right\|_\infty &\leq (1 - w_{k^*}) \max_k \|V_k - V_{k^*}\|_\infty \\ &\leq (K-1)e^{-\Delta_{\min}/T} C. \end{aligned}$$

Branch term. For a branch with margin $m = |z| \geq m_{\min}$, the gate error is $|\sigma(\alpha z) - \mathbb{I}[z > 0]| = \sigma(-\alpha m) \leq e^{-\alpha m_{\min}}$, so the blended counter $(1-g)\text{pc}_{\bar{b}} + g\text{pc}_b$ differs from the hard target by at most $|\text{pc}_b - \text{pc}_{\bar{b}}| e^{-\alpha m_{\min}} \leq C e^{-\alpha m_{\min}}$.

Per-step bound. Adding the two contributions gives, for every reachable x , $\delta := \|\Phi_S^T(x) - \Phi_H(x)\|_\infty \leq C[(K-1)e^{-\Delta_{\min}/T} + e^{-\alpha m_{\min}}]$.

Propagation. Let $L \geq 1$ be a Lipschitz constant of Φ_H in $\|\cdot\|_\infty$ and $e_t = \|x_t^{T,S} - x_t^H\|_\infty$ the trajectory error after t steps, with $e_0 = 0$. Each step adds a deviation of at most δ to the hard map applied to the accumulated error, giving $e_{t+1} \leq L e_t + \delta$; unrolling, $e_N \leq \delta \sum_{i=0}^{N-1} L^i = \delta (L^N - 1)/(L - 1)$. As $T \rightarrow 0$ (and $\alpha \rightarrow \infty$, since $m_{\min} = \Theta(1)$), $\delta = \mathcal{O}(e^{-\Delta_{\min}/T}) \rightarrow 0$, so the relaxed trajectory converges to the hard one and $\Phi_S^T \rightarrow \Phi_H$. $\square$

3 Implementation Effort

Figure 1 visualises the effort estimate reported in the main paper: 373 commits over a 29-day calendar span, of which the active sessions sum to about 137 hours.

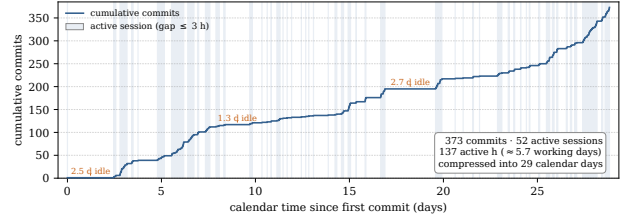


Figure 1: Implementation effort from the version-control log. Cumulative commits against calendar time; shaded bands are active sessions (consecutive commits no more than three hours apart). The long flat stretches are idle gaps. The 373 commits cluster into 52 active sessions totalling ~ 137 hours of work—about 5.7 working days—spread across a 29-day calendar window.